

EXPERIMENT-13

AIM: - Implement your own ping program

ALGORITHM

ping_program.py

```
import socket
import time

host = input("Enter host/IP address: ")

try:
    ip = socket.gethostbyname(host)
    print("Pinging", host, "[" + ip + "]")

    for i in range(4):
        start = time.time()

        try:
            s = socket.create_connection((ip, 80), timeout=2)
        except:
            print("Request timed out.")
            time.sleep(1)

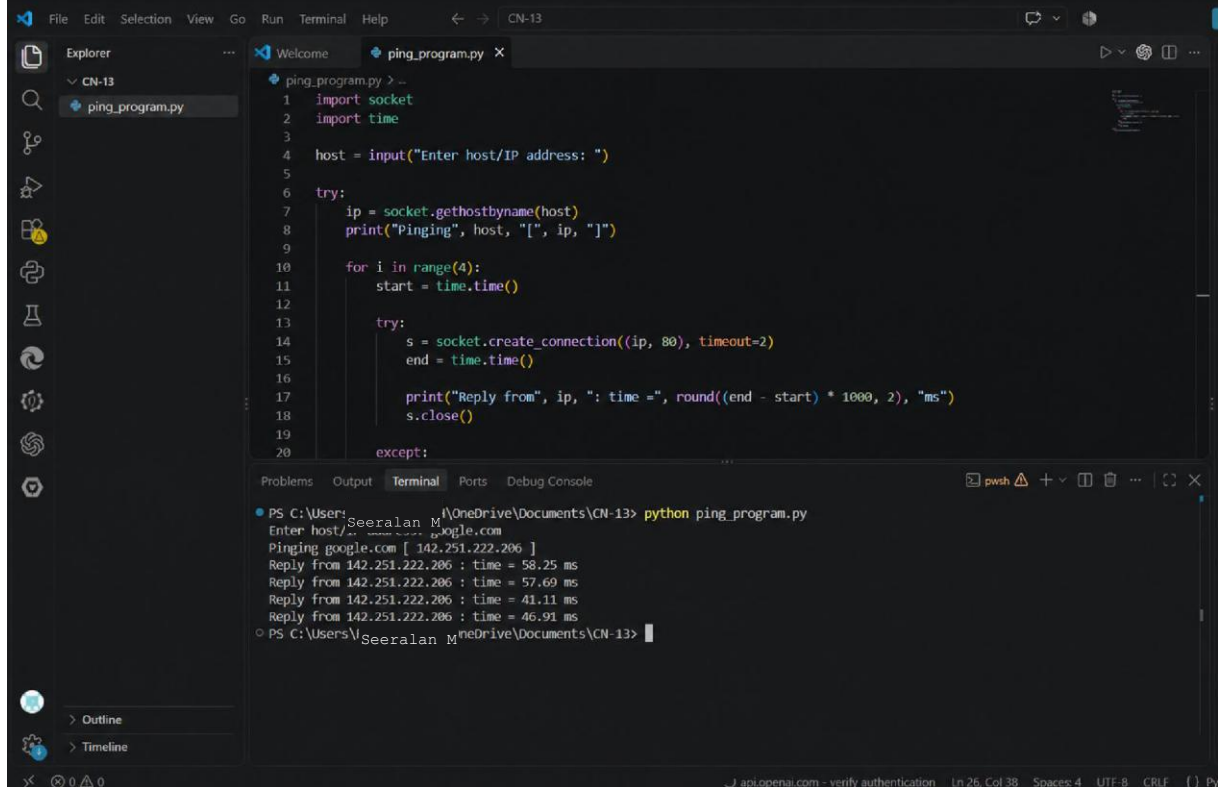
        end = time.time()

        print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")

    s.close()

except:
    print("Invalid host/IP address.")
```

```
● PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
○ PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> 
```



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a project named 'CN-13' with a file 'ping_program.py'. The main editor window displays the code for 'ping_program.py'. The code imports 'socket' and 'time', prompts the user for a host/IP address, and then performs a ping test using the 'socket' module. It prints the IP address and the response time for each of four attempts. The output window at the bottom shows the execution of the script, which matches the terminal output shown in the top section of the image.

```
1 import socket
2 import time
3
4 host = input("Enter host/IP address: ")
5
6 try:
7     ip = socket.gethostbyname(host)
8     print("Pinging", host, "[", ip, "]")
9
10    for i in range(4):
11        start = time.time()
12
13        try:
14            s = socket.create_connection((ip, 80), timeout=2)
15            end = time.time()
16
17            print("Reply from", ip, ": time =", round((end - start) * 1000, 2), "ms")
18            s.close()
19
20 except:
```

```
● PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> python ping_program.py
Enter host/IP address: google.com
Pinging google.com [ 142.251.222.206 ]
Reply from 142.251.222.206 : time = 58.25 ms
Reply from 142.251.222.206 : time = 57.69 ms
Reply from 142.251.222.206 : time = 41.11 ms
Reply from 142.251.222.206 : time = 46.91 ms
○ PS C:\Users\Seeralan M\OneDrive\Documents\CN-13> 
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